BHARATHRAJ SUBRAMANIAN

AI/ML Engineering Aspirant | Computer Science Undergraduate

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PROFESSIONAL SUMMARY

Computer Science undergraduate (CGPA 7.52) with hands-on experience in machine learning, computer vision and full-stack development, plus two internships. Built an end-to-end bus tracking platform spanning RFID hardware, a mobile app and a PostgreSQL backend. Seeking a Machine Learning Engineering internship.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C, C++

Machine Learning: PyTorch, TensorFlow, scikit-learn, OpenCV, HuggingFace Transformers, CUDA, Convolutional Neural Networks, Computer Vision

Web and Backend: React, React Native, Expo, FastAPI, Flask, Django, REST APIs, Supabase, PostgreSQL, PostGIS, Row Level Security, Tailwind CSS, Vite

Tools and Platforms: Git, GitHub, Linux, Raspberry Pi, NVIDIA Jetson Nano, SQLite, Docker (familiar)

EXPERIENCE

AI and ML Intern | HurryEP Technologies Pvt. Ltd. Jun 2026

- Two-week internship (8 to 24 June 2026) on applied machine learning tasks, working within a technical team to deliver assigned AI/ML project objectives on schedule.

Python Web Development Intern | Bread & Code Technology, Madurai Dec 2025 - Jan 2026

- One-month internship (5 December 2025 to 2 January 2026) in Python web development; rated Good by the founder for professional skills and dedication.

PROJECTS

College Bus Automation and Live Tracking System | Python, React, React Native, Supabase, PostgreSQL, Raspberry Pi 5

- Built a full-stack platform as an npm-workspaces monorepo (about 7,000 lines of TypeScript, Python and SQL): admin, student and gate portals in React, an Expo React Native driver app and a Python RFID gate service.
- Wrote a multithreaded Python service for a Raspberry Pi 5 that reads UHF RFID tags from a Chainway UR4 reader over TCP, infers bus entry or exit from two-antenna read order, suppresses duplicate reads, and logs to SQLite before syncing to Supabase so no gate event is lost when the network drops.
- Implemented background GPS tracking with Expo Location and Task Manager plus an offline queue; students view live bus position and Haversine-based ETA on Leaflet maps through Supabase Realtime.
- Designed a 13-table PostgreSQL schema with PostGIS and role-scoped Row Level Security; built passwordless student login (rate-limited Telegram OTP exchanged for a custom JWT in a Deno Edge Function) and phone OTP for drivers.
- Built a scheduled Edge Function for Telegram ETA alerts and late-arrival penalties; seeded with a Madurai college fleet of 23 buses, 241 schedules and about 890 students.

Other Projects

- **Diabetic Retinopathy Detector** (Python, CNN, OpenCV): Classifies retinopathy severity from retinal images using image preprocessing and CNN architectures.
- **Offline AI Coding Assistant** (CUDA): Locally run, GPU-accelerated coding assistant tuned for low-latency inference with no cloud dependency.
- **Facial Recognition and Remote Fingerprint Unlock** (Python, OpenCV LBPH, Tkinter): Windows authentication system combining face recognition with remote fingerprint unlock.
- **Aegis-X and EcoTrace** (React 18, TypeScript, Vite, JavaScript): Deployathon community dashboard, and a footprint calculator with charts and CSV, JSON and PDF export.

CERTIFICATIONS

Intro to Machine Learning | Kaggle	Jun 2026
Getting Started with AI on Jetson Nano | NVIDIA	Jun 2026
Computer Vision | Microsoft Azure	2026
AWS Certification Subject Matter Expert Training | AWS	Mar 2026
Python Developer Certification, 300 hours | freeCodeCamp	Dec 2025
The Joy of Computing using Python, Elite (70%) | NPTEL, IIT Madras	Jan - Apr 2025

EDUCATION

B.E. Computer Science and Engineering | SRM Madurai College for Engineering and Technology | CGPA 7.52 Expected May 2028